65. *The marks of 1000 students in an examination follows a normal distribution with mean 70 and standard deviation 5. Find the number of students whose marks will be*
(i) less than 65 (ii) more than 75 (iii) between 65 and 75

>> Let x represent the marks of students.

By data $\mu = 70,\ \sigma = 5$ Hence s.n.v $z = \dfrac{x-\mu}{\sigma} = \dfrac{x-70}{5}$

(i) If $x = 65,\ z = -1$ and we have to find $P(z < -1)$

$$P(z < -1) = P(z > 1)$$
$$= P(z \geq 0) - P(0 < z < 1)$$
$$= 0.5 - \phi(1) = 0.5 - 0.3413 = 0.1587$$

$\therefore$ **number of students scoring less than 65 marks**

$$= 1000 \times 0.1587 = 158.7 \approx \mathbf{159}$$

(ii) If $x = 75,\ z = 1$ and we have to find $P(z > 1)$

$$P(z > 1) = P(z \geq 0) - P(0 < z < 1)$$
$$= 0.5 - \phi(1) = 0.5 - 0.3413 = 0.1587$$

$\therefore$ **number of students scoring more than 75 marks**

$$= 1000 \times 0.1587 = 158.7 \approx \mathbf{159}$$

(iii) We have to find $P(-1 < z < 1)$

$$P(-1 < z < 1) = 2P(0 < z < 1)$$
$$= 2\,\phi(1) = 2(0.3413) = 0.6826$$

$\therefore$ **number of students scoring marks between 65 and 75**

$$= 1000 \times 0.6826 = 682.6 \approx \mathbf{683}$$

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66. *In a test on electric bulbs, it was found that the life time of a particular brand was distributed normally with an average life of 2000 hours and S.D of 60 hours. If a firm purchases 2500 bulbs find the number of bulbs that are likely to last for (i) more than 2100 hours. (ii) less than 1950 hours (iii) between 1900 to 2100 hours.*

>> By data $\mu = 2000,\ \sigma = 60$

We have s.n.v $z = \dfrac{x-\mu}{\sigma} = \dfrac{x-2000}{60}$

(i) To find $P(x > 2100)$

If $x = 2100,\ z = 100/60 \approx 1.67$

$P(x > 2100) = P(z > 1.67)$

$= P(z \geq 0) - P(0 < z < 1.67)$

$= 0.5 - \phi(1.67)$

$= 0.5 - 0.4525 = 0.0475$

...mber of bulbs that are likely to last for more than 2100 hours is

$2500 \times 0.0475 = 118.75 \approx$ **119**

...o find $P(x < 1950)$

...f $x = 1950$, $z = -5/6 = -0.83$

$P(x < 1950) = P(z < -0.83)$

$= P(z > 0.83)$

$= P(z \geq 0) - P(0 < z < 0.83)$

$= 0.5 - \phi(0.83)$

$= 0.5 - 0.2967 = 0.2033$

...umber of bulbs that are likely to last for less than 1950 hours is

$2500 \times 0.2033 = 508.25 \approx$ **508**

To find $P(1900 < x < 2100)$

If $x = 1900$, $z = -1.67$ and if $x = 2100$, $z = 1.67$

$P(1900 < x < 2100) = P(-1.67 < z < 1.67)$

$= 2P(0 < z < 1.67)$

$= 2\phi(1.67) = 2 \times 0.4525 = 0.905$

number of bulbs that are likely to last between 1900 and 2100 hours

$= 2500 \times 0.905 = 2262.5 \approx$ **2263**

In a normal distribution 31% of the items are under 45 and 8% of the items are over 64. Find the mean and S.D of the distribution.

Let μ and σ be the mean and S.D of the normal distribution

...data $P(x < 45) = 0.31$ and $P(x > 64) = 0.08$

We have s.n.v $z = \dfrac{x - \mu}{\sigma}$

When $x = 45$, $z = \dfrac{45 - \mu}{\sigma} = z_1$ *(say)*

$x = 64$, $z = \dfrac{64 - \mu}{\sigma} = z_2$ *(say)*

Thus we have,

$$P(z < z_1) = 0.31 \text{ and } P(z > z_2) = 0.08$$

i.e., $0.5 + \phi(z_1) = 0.31$ and $0.5 - \phi(z_2) = 0.08$

$\Rightarrow$ $\phi(z_1) = -0.19$ and $\phi(z_2) = 0.42$

Refering to the normal probability tables we have

$$0.1915\,(\approx 0.19) = \phi(0.5) \text{ and } 0.4192\,(\approx 0.42) = \phi(1.4)$$

$\therefore$ $\phi(z_1) = -\phi(0.5)$ and $\phi(z_2) = \phi(1.4)$

$\Rightarrow$ $z_1 = -0.5$ and $z_2 = 1.4$

i.e., $\dfrac{45-\mu}{\sigma} = -0.5$ and $\dfrac{64-\mu}{\sigma} = 1.4$

or $\mu - 0.5\,\sigma = 45$ and $\mu + 1.4\,\sigma = 64$

By solving we get $\mu = 50,\ \sigma = 10$

Thus **mean = 50 and S.D = 10**

68. *In an examination 7% of students score less than 35% marks and 89% of students score less than 60% marks. Find the mean and standard deviation if the marks are normally distributed. It is given that if*

$$P(z) = \frac{1}{\sqrt{2\pi}} \int_0^z e^{-z^2/2}\,dz \quad \textit{then} \quad P(1.2263) = 0.39 \textit{ and } P(1.4757) = 0.[illegible]$$

>> Let μ and σ be the mean and S.D of the normal distribution.

By data we have $P(x < 35) = 0.07$, $P(x < 60) = 0.89$

We have s.n.v $z = \dfrac{x-\mu}{\sigma}$

When $x = 35$, $z = \dfrac{35-\mu}{\sigma} = z_1$ (*say*)

$x = 60$, $z = \dfrac{60-\mu}{\sigma} = z_2$ (*say*)

Hence we have

$$P(z < z_1) = 0.07 \text{ and } P(z < z_2) = 0.89$$

i.e., $0.5 + \phi(z_1) = 0.07$ and $0.5 + \phi(z_2) = 0.89$

$\therefore$ $\phi(z_1) = -0.43$ and $\phi(z_2) = 0.39$

...g the given data in the RHS of these we have,

$$\phi(z_1) = -\phi(1.4757) \text{ and } \phi(z_2) = \phi(1.2263)$$

$$z_1 = -1.4757 \quad \text{and} \quad z_2 = 1.2263$$

$$\frac{35-\mu}{\sigma} = -1.4757 \quad \text{and} \quad \frac{60-\mu}{\sigma} = 1.2263$$

$$\mu - 1.4757\,\sigma = 35 \text{ and } \mu + 1.2263\,\sigma = 60$$

...olving we get $\mu = 48.65$ and $\sigma = 9.25$

...s **mean = 48.65 and S.D = 9.25**

For the following normal distribution find c and also the mean and standard deviation of the frequency distribution.

$$f(x) = c\,e^{\frac{-1}{24}(x^2-6x+4)}$$

We have the p.d.f of the normal distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\,e^{-(x-\mu)^2/2\sigma^2} \qquad \ldots(1)$$

...nsider $f(x) = c\,e^{\frac{-1}{24}(x^2-6x+4)}$

$$f(x) = c\,e^{\frac{-1}{24}[(x-3)^2-5]} = c\;e^{5/24}\;e^{-(x-3)^2/24}$$

$$f(x) = c\;e^{5/24}\;e^{-(x-3)^2/2(\sqrt{12})^2} \qquad \ldots(2)$$

...mparing the exponents of the exponential function in (1) and (2) we have $\mu = 3$ and ... $= \sqrt{12}$ provided

$$c\,e^{5/24} = \frac{1}{\sigma\sqrt{2\pi}} = \frac{1}{\sqrt{12}\,\sqrt{2\pi}} = \frac{1}{\sqrt{24\pi}}$$

...hus $c = \dfrac{e^{-5/24}}{\sqrt{24\pi}}$; **Mean $(\mu) = 3$, S.D$(\sigma) = \sqrt{12}$**